ANSWER KEY & MARKING SCHEME SET 1

Chapter 7: **How do Organisms Reproduce?** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** Offspring produced by budding in Hydra	Asexual offspring are clones with minor copying errors.
2	**(a)** filament	The anther stalk.
3	**(d)** 23	Gametes have half the number.
4	**(c)** Plasmodium	Multiple fission under unfavourable conditions.
5	**(b)** unisexual (male)	Only male parts are present.
6	**(a)** 46 chromosomes	Sperm (23) + egg (23).
7	**(c)** Inaccurate DNA copying produces variation	Errors in copying provide the basis of variation.
8	**(c)** 20 seeds	Each fertilised ovule forms one seed.
9	**(d)** umbilical cord	It carries nutrients to the foetus.
10	**(a)** vegetative propagation	Buds in the eyes sprout.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Unicellular organisms are a single cell and can split (1); Complex organisms have organ systems and need specialised reproductive organs (1).	1+1
12	Only one parent copies its DNA (1); Only small copying errors occur, so offspring are almost identical (1).	1+1
13	A sticky surface holds the pollen (1); It increases the chance of pollen capture (1).	1+1
14	Spores of Rhizopus fall on it and germinate (1); Hyphae spread rapidly and form many sporangia (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Spores are tiny and light, so they are dispersed far (1); Many do not find moisture and food (1); Large numbers ensure survival of some (1).	1+1+1
16	Male gametes are produced in large numbers (1); They are small and motile and many are lost before reaching the egg (1); Female gametes are few, larger and rich in food (1).	1+1+1
17	Quick multiplication of plants (1); Seedless plants like banana and orange can be grown (1); Desirable traits are maintained (1).	1+1+1
	OR: Vegetative propagation by suckers (1); Offspring have the same genes as the parent (1); Plants also bear fruits earlier (1).	1+1+1
18	Sperms have to travel to reach the egg (1); Tail of the sperm helps swimming (1); Egg is large and non-motile, holding stored food (1).	1+1+1
	OR: Variation provides survival chances under changed conditions (1); Example: a bacterium resistant to heat survives a heat wave (1); The species continues through such variants (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Binary fission in Amoeba and multiple fission in Plasmodium (1); Budding in yeast and Hydra (1); Fragmentation in Spirogyra and regeneration in Planaria (1); Spore formation in Rhizopus (1); Complex animals have specialised organs which cannot be rebuilt from fragments (1).	1+1+1+1+1
	OR: Ovaries produce eggs and hormones (1); Oviduct is the site of fertilisation (1); Uterus receives and nourishes the embryo (1); Placenta exchanges materials with mother's blood (1); Cervix and vagina form the birth canal (1).	1+1+1+1+1
20	Bryophyllum leaf notches grow plantlets (1); Rose cutting grows roots and shoots (1); Potato eye sprouts into a new plant (1); All are vegetative propagation (1); New plants are genetically identical to the parent (1).	1+1+1+1+1
	OR: An egg is released from the ovary (1); Uterine lining thickens (1); (a) Zygote implants and the placenta forms (1); (b) The lining breaks and menstruation occurs (1); The cycle begins again (1).	1+1+1+1+1

Mistakes to avoid:

- Writing that Plasmodium reproduces by binary fission; it undergoes **multiple** fission, whereas Amoeba shows binary fission.
- Using 'fertilisation' for pollination; pollination is transfer of pollen, fertilisation is fusion of male and female gametes.
- Saying that fertilisation takes place in the uterus; it occurs in the oviduct and the zygote is implanted in the uterus.
- Confusing regeneration with reproduction by fragmentation; regeneration uses specialised cells to rebuild a missing part, e.g. Planaria, Hydra.
- Stating that asexual reproduction produces absolutely identical offspring; tiny variations are produced by errors in DNA copying.
- Believing that the testes lie inside the abdomen; they are in the scrotum because sperm formation requires a lower temperature.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 7: **How do Organisms Reproduce?** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** self-pollination	Same plant, so genetically it counts as self-pollination.
2	**(a)** In the urethra / vas deferens region	Semen is formed before ejaculation.
3	**(c)** they survive unfavourable conditions	The tough coat protects the spore.
4	**(b)** pollination, pollen tube growth, fertilisation, fruit formation	Sequence of events in a flower.
5	**(a)** Testosterone	Testes secrete testosterone.
6	**(a)** the damp dark place	Moisture favours fungal growth.
7	**(d)** Cotyledon	Stored food is in the cotyledons.
8	**(c)** ovaries	Ovaries secrete the female hormone.
9	**(c)** grow into a new filament	Fragmentation.
10	**(d)** it lacks water	Water activates enzymes.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Banana is seedless (1); Plants bear fruits earlier and are identical to the parent (1).	1+1
12	Sperm production would fall (1); Body temperature is too high for sperm formation (1).	1+1
13	Each piece grows into a complete Planaria (1); Regeneration (1).	1+1
14	It brings together genes of two plants (1); The resulting variation helps the species adapt (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Each gamete has half the number of chromosomes (1); Fertilisation restores the full number in the zygote (1); So the number is the same in every generation (1).	1+1+1
16	Testis, epididymis, vas deferens (1); Urethra, penis, vagina of the female (1); Through the uterus into the oviduct where it meets the egg (1).	1+1+1
17	Hydra has cells that can specialise into various tissues (1); Humans have a complex body organisation (1); Human cells cannot regrow the whole limb (1).	1+1+1
	OR: Pollen on the stigma develops a tube (1); The tube grows through the style to the ovule (1); Male gamete travels through it to fuse with the egg (1).	1+1+1
18	Embedding of the embryo in the uterine lining (1); The embryo gets nutrients and oxygen from the mother's blood (1); Without it the pregnancy cannot continue (1).	1+1+1
	OR: Pollen carries the male gametes (1); Ovule has the female gamete (1); Their fusion produces the zygote that makes the seed (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Sepals and petals (1); Stamen with anther and filament (1); Pistil with stigma, style and ovary (1); Ovary contains ovules (1); Pollen tube travels stigma, style, ovary and enters the ovule (1).	1+1+1+1+1
	OR: Parents: one or two (1); Variation: low or high (1); Speed: fast or slower (1); Gametes: absent or present (1); Examples: budding in Hydra, humans and flowering plants (1).	1+1+1+1+1
20	Pollen grain lands on the stigma (1); Pollen tube grows down the style (1); Male gamete fuses with the egg in the ovule (1); Zygote becomes the embryo; ovule becomes the seed (1); Ovary wall becomes the fruit; other parts wither (1).	1+1+1+1+1
	OR: Sperm fuses with egg in the oviduct (1); Zygote divides to form an embryo (1); Embryo is implanted in the uterus lining (1); Placenta has villi and brings blood close for exchange (1); Nutrients and oxygen pass to the foetus; wastes pass back (1).	1+1+1+1+1

Mistakes to avoid:

- Writing that Plasmodium reproduces by binary fission; it undergoes **multiple** fission, whereas Amoeba shows binary fission.
- Using 'fertilisation' for pollination; pollination is transfer of pollen, fertilisation is fusion of male and female gametes.
- Saying that fertilisation takes place in the uterus; it occurs in the oviduct and the zygote is implanted in the uterus.
- Confusing regeneration with reproduction by fragmentation; regeneration uses specialised cells to rebuild a missing part, e.g. Planaria, Hydra.
- Stating that asexual reproduction produces absolutely identical offspring; tiny variations are produced by errors in DNA copying.
- Believing that the testes lie inside the abdomen; they are in the scrotum because sperm formation requires a lower temperature.

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MASTERMIND

ANSWER KEY & MARKING SCHEME SET 3

Chapter 7: **How do Organisms Reproduce?** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** implantation of the embryo	The embryo implants here.
2	**(b)** Hydra - budding	Hydra reproduces by buds.
3	**(c)** Ovary wall	The ovary wall becomes the pericarp.
4	**(c)** does not occur	The lining is retained.
5	**(a)** Human beings	Complex organisms cannot regenerate whole bodies.
6	**(d)** pedicel	The pedicel carries the flower.
7	**(a)** ovulation	Ovulation.
8	**(a)** Binary fission	Binary fission is asexual.
9	**(a)** Much pollen is wasted before reaching a stigma	Chances of pollination increase.
10	**(b)** Breast development	Girls develop breasts at puberty.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Embryo would not receive nutrients and oxygen (1); Wastes would not be removed and it cannot survive (1).	1+1
12	Yeast is a single cell and a bud is a bulge of the cell (1); Hydra bud is a multicellular outgrowth forming from the body wall (1).	1+1
13	Ovule becomes the seed (1); Ovary becomes the fruit (1).	1+1
14	To prepare for the possible implantation of the embryo (1); The lining is richly supplied with blood vessels for nourishment (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Seed develops from the fertilised ovule (1); Fruit develops from the ovary (1); A fruit encloses the seed and helps in its dispersal (1).	1+1+1
16	Placenta has villi to exchange materials (1); Umbilical cord links the embryo and placenta (1); Nutrients and oxygen reach the foetus and wastes leave (1).	1+1+1
17	It stretches to hold the growing foetus (1); Muscles contract at birth to push the baby out (1); Thick lining supports implantation (1).	1+1+1
	OR: Unisexual flowers have either stamens or pistil (1); Bisexual flowers have both (1); Papaya (unisexual), mustard or hibiscus (bisexual) (1).	1+1+1
18	Testosterone causes beard growth and voice deepening in boys (1); Oestrogen causes breast development and menstruation in girls (1); Both control the maturation of reproductive organs (1).	1+1+1
	OR: Both produce gametes (1); Both secrete sex hormones (1); Ovary produces eggs and oestrogen; testis produces sperms and testosterone (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Cells make copies of their DNA (1); DNA copying uses chemical reactions that are not perfectly accurate (1); Small errors give variations (1); Variations help some individuals survive changes (1); Without variation a population may become extinct (1).	1+1+1+1+1
	OR: Boys: voice deepens, beard grows (1); Girls: breast growth, menstruation begins (1); Hair appears in underarms and genital area (1); Testosterone in boys and oestrogen in girls (1); Reproductive organs mature and begin to function (1).	1+1+1+1+1

20	Testis produces sperms and testosterone (1); Scrotum keeps testes at a lower temperature (1); Vas deferens carries sperms to the urethra (1); Prostate and seminal vesicles provide fluid (1); Penis and urethra transfer sperms into the female body (1).	1+1+1+1+1
	OR: Seed coat, cotyledons and embryo (1); Plumule and radicle form the embryo axis (1); Water, air and suitable temperature needed (1); Radicle grows down, plumule up (1); Cotyledons give food until leaves can make it (1).	1+1+1+1+1

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